Reacting Masses

Exam code: 0620 & 0971

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Reacting masses

Extended tier only

- Chemical / symbol equations can be used to calculate:
  - The **moles** of reactants and products
  - The **mass** of reactants and products
- To do this:
  - Information from the question is used to find the amount in moles of the substances being considered
  - Then, the **ratio** between the substances is identified using the balanced chemical equation
  - Once the moles have been determined they can then be converted into grams using the relative atomic or relative formula masses

Worked Example

Magnesium undergoes combustion to produce magnesium oxide.

The overall reaction that is taking place is shown in the equation below.

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

completely burning 6.0 g of magnesium in oxygen in the following reaction:

$$A_r(O) = 16 \quad A_r(Mg) = 24$$

Answer:

1. Calculate the moles of magnesium
   - Moles = $\frac{\text{mass}}{M_r} = \frac{6}{24} = 0.25$
2. Use the molar ratio from the balanced symbol equation
   - 2 moles of magnesium produce 2 moles of magnesium oxide
   - The ratio is 1 : 1
   - Therefore, 0.25 moles of magnesium oxide is produced
3. Calculate the mass of magnesium oxide
   - Mass = moles x M_r = 0.25 moles x (24 + 16) = 10 g

Worked Example

In theory, aluminium could decompose as shown in the equation below.

$$2Al_2O_3 \rightarrow 4Al + 3O_2$$

Calculate the maximum possible mass of aluminium, in tonnes, that can be produced from 51 tonnes of aluminium oxide.

$$A_r(O) = 16 \quad A_r(Al) = 27$$

Answer:

1. Calculate the moles of aluminium oxide
   - Mass = 51 tonnes x 10^6 = 51 000 000 g
   - M_r of aluminium oxide = (2 x 27) + (3 x 16) = 102
   - Moles = $\frac{\text{mass}}{M_r} = \frac{51\ 000\ 000}{102} = 500\ 000$
2. Use the molar ratio from the balanced symbol equation
   - 2 moles of aluminium oxide produces 4 moles of aluminium
   - The ratio is 1 : 2
   - Therefore, 2 x 500 000 = 1 000 000 moles of aluminium is produced
3. Calculate the mass of aluminium
   - Mass = moles x M_r = 1 000 000 moles x 27 = 27 000 000 g
   - Mass in tonnes = $\frac{27\ 000\ 000}{10^6}$ = 27 tonnes

Examiner Tips and Tricks

Remember: The molar ratio of a balanced equation gives you the ratio of the amounts of each substance in the reaction.

- A chemical reaction stops when one of the reactants is used up
- The reactant that is used up first is the **limiting reactant**, as it limits the duration and hence the amount of product that a reaction can produce
  - The one that is remaining is the **excess** reactant
  - The limiting reagent is the reactant which is **not present in excess** in a reaction
- The amount of product is therefore **directly proportional** to the amount of the limiting reactant added at the beginning of a reaction

Determining the limiting reactant

- In order to determine which reactant is the limiting reagent in a reaction, we have to consider the amounts of each reactant used and the molar ratio of the balanced chemical equation
- When performing reacting mass calculations, the limiting reagent is always the number that should be used, as it indicates the maximum possible amount of product that can form
  - Once all of a limiting reagent has been used up, the reaction cannot continue
- The steps are:
  1. Convert the mass of each reactant into moles by dividing by the molar masses
  2. Write the balanced equation and determine the molar ratio
  3. Look at the equation and compare the moles

Worked Example

9.2 g of sodium is reacted with 8.0 g of sulfur to produce sodium sulfide, Na_2S.

Which reactant is in excess and which is the limiting reactant?

Relative atomic masses (A_r): Na = 23; S = 32

Answer:

1. Calculate the moles of each reactant
   - Moles = $\frac{\text{mass}}{M_r}$
   - Moles Na = $\frac{9.2}{23}$ = 0.40
   - Moles S = $\frac{8.0}{32}$ = 0.25
2. Write the balanced equation and determine the molar ratio
   - $2Na + S \rightarrow Na_2S$
   - So, the molar ratio of Na : S is 2 : 1
3. Compare the moles

- S is in excess
- Therefore, Na is the limiting reactant

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